

FAST Infinity: Campus Life Simulator

Group Members

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Abstract

FAST Survival Challenge is an interactive campus-based simulation system where a student character navigates through a virtual model of FAST-NUCES and interacts with different university facilities. The system integrates a Cafeteria Management System, Bookshop Management System, and an E-Sports Gaming Center. Scores earned from games are converted into virtual currency, which can be used to purchase items within the campus, demonstrating a fully integrated database-driven transactional system.

Description

FAST Survival Challenge is a comprehensive database-driven system designed to simulate essential campus operations within a virtual university environment. The system connects multiple modules into one unified platform:

1. Campus Simulation Interface

An interactive model of FAST-NUCES campus where the user (student character) can navigate and enter different facilities.

2. E-Sports Gaming Center

- Contains multiple games
- Each game generates a score.
- The score is converted into virtual cash and stored in the database.
- Leaderboards and game history are maintained.

3. Cafeteria Management System

- Displays menu items.
- Allows purchase of food items using earned virtual cash.
- Manages stock levels and inventory updates.
- Stores purchase history and transaction logs.

4. Bookshop Management System

- Displays books and other products.
- Handles item purchasing and stock management.
- Tracks user purchases and generates receipts.

5. Wallet & Transaction System

- Each user has a wallet.
- Game scores are converted into wallet balance.
- Purchases deduct balance through secure database transactions.
- All activities are logged for auditing and reporting.

Key Functionalities

- User registration and login
- Wallet balance management
- Game score recording and currency conversion
- Item purchase with stock validation
- Order history tracking
- Leaderboard generation
- Inventory monitoring
- Transaction logging

The project demonstrates normalization, relational integrity, transaction management (ACID properties), triggers, stored procedures, and advanced SQL queries.

List of Tables to be Implemented (Minimum Core Tables)

1. Users

Stores user account details.

- user_id (PK)
 - username
 - email
 - password
 - created_at
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2. Wallet

Stores user balance.

- user_id (PK, FK → Users)
 - balance
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3. Games

Stores game information.

- game_id (PK)
 - game_name
 - max_score
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4. Game_Sessions

Stores each gameplay record.

- session_id (PK)
- user_id (FK → Users)
- game_id (FK → Games)
- score
- earned_cash

- played_at
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5. Items

Stores all cafeteria and bookshop items.

- item_id (PK)
 - item_name
 - category (Cafeteria / Bookshop)
 - price
 - stock
-

6. Orders

Stores purchase orders.

- order_id (PK)
 - user_id (FK → Users)
 - total_amount
 - order_date
-

7. Order_Details

Stores items inside each order.

- order_detail_id (PK)
 - order_id (FK → Orders)
 - item_id (FK → Items)
 - quantity
 - price
-

8. Transaction_Log

Stores all wallet transactions.

- transaction_id (PK)
 - user_id (FK → Users)
 - amount
 - transaction_type (Credit/Debit)
 - timestamp
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Conclusion

This project integrates gaming, campus services, and financial management into a unified database-driven system. It not only demonstrates strong relational database concepts and normalization but also provides a practical, interactive system that simulates real-world transactional operations within a university environment.